Kernel Development

1. Using bsp

bsp is a set of tools provided by Radxa to quickly build U-Boot Kernels. It is very easy to build your own U-Boot and Kernel with bsp.

The following shows you how to build a kernel with bsp using Radxa ROCK 5B as an example.

1.1 bsp environment configuration

The following is the recommended environment for `bsp`.
Hardware: x86_64 PC recommended
System: Ubuntu 22.04 or Debian 12

Install dependencies

```
sudo apt update
sudo apt install -y git qemu-user-static binfmt-support

# Podman (recommended)
sudo apt install -y podman podman-docker
sudo touch /etc/containers/nodocker
# Docker
# sudo apt install docker.io

# Optional dependencies for minor features
sudo apt install -y systemd-container
```

Getting the code

`bsp` uses the git submodule. Therefore, use the following command to get the code:

```
git clone --recurse-submodules https://github.com/radxa-repo/bsp.git
```

Understanding `bsp` usage

Once you have the `bsp` code on your machine, you can run `bsp` to see the built-in help:

```
cd bsp
./bsp
```

1.2 Build the kernel

Running `./bsp` will show you all the arguments and their explanations, and the command `bsp [options] <linux|u-boot> <profile> [product]` will be available at the prompt.
Taking Radxa ROCK 5B as an example, the linux profile is "rk2410". The following are profiles for other products:

product	profile	product	profile	product	profile	product	profile
rock-4se	rockchip	radxa-cm3-io	rk356x	radxa-zero	stable	rock-5a	rk2410
rock-pi-4a-plus	rockchip	radxa-cm3-rpi-cm4-io	rk356x	radxa-zero-2pro	stable	rock-5b	rk2410
rock-pi-4b-plus	rockchip	radxa-cm3-sodimm-io	rk356x	rock-s0	stable	rock-5b-plus	rk2410
rock-4c-plus	rockchip	radxa-cm3i-io	rk356x	rock-pi-s	stable	rock-5c	rk2410
rock-pi-4a	rockchip	radxa-cm3j-rpi-cm4-io	rk356x	rock-pi-n10	stable	rock-5d	rk2410
rock-pi-4b	rockchip	radxa-e23	rk356x	rock-pi-e	stable	rock-5t	rk2410
rock-pi-4c	rockchip	radxa-e25	rk356x			rock-5-itx	rk2410

product	profile	product	profile	product	profile	product	profile
rock-4-core-io	rockchip	radxa-zero3	rk356x			radxa-e52c	rk2410
		rock-3a	rk356x			radxa-e54c	rk2410
		rock-3b	rk356x			radxa-nx5-io	rk2410
		rock-3c	rk356x			radxa-cm4-rpi-cm4-io	rk2410
		rock-2	rk356x			radxa-cm5-io	rk2410
						radxa-cm5-rpi-cm4-io	rk2410
						radxa-rk3576	rk2410

TIP

If the product model is not found in the table, select the corresponding option based on your product's SoC model.

Example: Radxa CM4 uses the Rockchip RK3576 SoC. According to the table, select `rk2410` as the `profile` parameter for `radxa-rk3576`.

Then fill in the `profile` parameter of the `bsp` command with the product's corresponding `profile`: `rk2410` to pull and configure the code, for example:

```
./bsp linux rk2410 --no-build
# `--no-build` only configure the code without compile
```

The path to the kernel source code is located in `.src/linux` in the bsp directory, and the kernel can be built again after modifying the kernel source code:

```
cd bsp
./bsp --no-prepare-source linux rk2410 -r 999
# Parameter description:
# `--no-prepare-source` # This parameter is not needed for first time compilation, it is added to compile with local modifications, if this parameter is not added, the latest code will be synchronized from the Radxa kernel repository and overwrite the local modifications.
# `-r 999` # Specify the kernel version as 999 for priority use.
```

After compilation, a number of `deb` packages will be generated in the current directory, you only need to install the following two `deb` packages

> **INFO**
>
> The software package file name will change according to the parameters of `propile` and `-r`

```
linux-headers-6.1.84-999-rk2410_6.1.84-999_arm64.deb
linux-image-6.1.84-999-rk2410_6.1.84-999_arm64.deb
```

Copy the above two `deb` packages to the board and install them with the `dpkg` command to complete the kernel installation

```
sudo dpkg -i linux-headers-6.1.84-999-rk2410_6.1.84-999_arm64.deb
sudo dpkg -i linux-image-6.1.84-999-rk2410_6.1.84-999_arm64.deb
sudo reboot
```

After the reboot is complete, you can check if the installation was successful by checking the version number of the currently booted kernel with `uname -a`.

2. Compile manually

> **TIP**
>
> `bsp` is the only compilation method currently being maintained, and it makes additional changes to the source code when compiling. If you want to develop based on our

officially released software, you should at least use `bsp` to download and configure the code.

If you want to build with a custom `make` command, please refer to the Prepare working tree section to configure the code.

3. Documentation

- bsp official documentation

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